THE BEGINNER'S GUIDE TO

Caravan Solar

Clear recommendations, brand comparisons, and shopping lists for Australian caravanners who don't want to become electrical engineers.

Sized for how you travel. Weekender, Tourer, or Full-timer.

Weekender

Tourer

Full-timer

PDF GUIDE + WORKSHEETS

$49

SECTION 1

The 60-Second Version

SKIP THIS IF YOU WANT

If you just want the "what to buy" part, go straight to Section 3. But if you want the 60-second version of what you're putting in your caravan, here it is.

The Whole System in Plain English

1. Solar panels sit on your roof and turn sunlight into power.
2. A battery stores that power. This is the fuel tank.
3. A charge controller sits between panels and battery to manage charging safely.
4. A DC-DC charger tops up the battery from your car while you drive.
5. An inverter lets you plug in normal household things (laptop, microwave).
6. A battery monitor is your fuel gauge. It tells you how much power is left.

That's it. Six things. The rest of this guide tells you exactly which ones to buy and what size.

One Quick Concept

You'll see the word watt-hours a lot. It's how we measure daily power use. If something uses 60 watts and runs for 10 hours, that's 600 watt-hours. Your fridge uses about 60W and runs all day, so roughly 1,440 Wh per day.

That daily number drives everything. The battery size, the amount of solar, the whole lot. Section 2 helps you work out yours. It takes about 10 minutes.

Your Power Audit

We need to figure out roughly how much power your caravan uses in a day. This takes about 10 minutes, and it's the one piece of homework in this entire guide. Everything else flows from it.

Don't stress about being exact. Within 20% is plenty. We add a buffer for the stuff you forget.

Walk-Through: Dave and Sarah's Setup

Dave and Sarah are tourers. Weeks on the road, mix of caravan parks and free camping. Here's their typical day off-grid:

Appliance	Usage	Daily Wh
Fridge (all day, all night)	60W x 24h	~1,440 Wh
LED lights (evening)	30W x 5h	~150 Wh
TV after dinner	40W x 3h	~120 Wh
Phones + laptop	100W x 2h	~220 Wh
Water pump, heater, etc.	Various	~220 Wh

Daily total: about 2,150 Wh. Add 25% buffer: about 2,700 Wh per day.

That's it. No complicated formulas. Just: what do you have, how long is it on, roughly how much does it use?

Your turn. Use the Power Audit Worksheet at the back. It lists every common caravan appliance with typical power draws.

Which Tier Are You?

Your daily number tells you what kind of system you need. Pick the one that fits:

WEEKENDER — $1,500 - $2,500

Under 1,500 Wh/day

Short trips, mostly caravan parks, some free camping. Fridge, lights, phones.

TOURER — $3,000 - $5,000

1,500 to 3,000 Wh/day

Weeks on the road, mix of parks and free camping. Fridge, lights, TV, laptops.

FULL-TIMER — $5,000 - $8,000+

Over 3,000 Wh/day

Living on the road. Everything above plus microwave, coffee machine, hair dryer.

A NOTE ON AIR CONDITIONING

Running air con off-grid on a standard setup isn't practical. It uses more power than everything else combined. If that's a must-have, you're looking at a specialised 48V system ($10,000+) or a generator. This guide covers standard 12V setups.

Got your number? Good. From here on, the guide is built around your tier. Every recommendation, every shopping list is matched to Weekender, Tourer, or Full-timer.

WORKSHEET

Power Audit

List every electrical appliance you plan to run off-grid. Check the label for wattage. If it shows amps, multiply by 12 to get watts.

Appliance	Watts	Hrs/Day	Daily Wh
Compressor fridge	50-70	24	____________
LED lights (all)	20-40	____	____________
TV (12V)	30-60	____	____________
Phone charger x ___	10-20 ea	____	____________
Laptop charger	45-65	____	____________
Tablet charger	10-15	____	____________
Water pump	50-70	____	____________
Diesel heater fan	20-40	____	____________
Range hood fan	20-30	____	____________
Radio/speakers	10-20	____	____________
CPAP machine	30-60	____	____________
Electric blanket	40-60	____	____________

Power Audit Worksheet

Appliance	Watts	Hrs/Day	Daily Wh
Hair dryer (inverter)	1200-2000	____	____________
Microwave (inverter)	800-1200	____	____________
Coffee machine (inv.)	800-1500	____	____________
Toaster (inverter)	800-1200	____	____________
______________	_____	____	____________
______________	_____	____	____________
______________	_____	____	____________

Subtotal: _______________ Wh

+ 25% buffer (x 1.25): _______________ Wh

Your Traveller Tier

Under 1,500 Wh	1,500-3,000 Wh	Over 3,000 Wh
Weekender	**Tourer**	**Full-timer**

My tier: ________________________

My daily total: ______________ Wh